

Harrison Tidy

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EDUCATION

University of British Columbia

Bachelor of Applied Science - Computer Engineering | GPA: 4.0/4.0

Vancouver, BC

Expected Apr 2029

WORK EXPERIENCE

WriteAway AI | Founder & Head of Engineering

2024 - Present

- Founded WriteAway to bring AI assistance into a compact handheld pen; interviewed and hired engineers and currently lead a 4-person team developing the product.
- Helped select chips and shape the system architecture; worked with engineers to debug embedded software and hardware integration problems.
- Led hands-on prototyping and debugging across nearly 100 mechanical prints and 5+ PCB revisions, helping resolve issues between firmware, electronics, and mechanical parts.

Creation Technologies

Jun 2026 - Aug 2026

PCB Assembly & Soldering Technician

Burnaby, BC

- Assembled, hand-soldered, and inspected PCB assemblies for military, industrial, and medical applications while following ESD, workmanship, and production quality procedures.
- Used microscopes and production inspection equipment to identify soldering, component-placement, and assembly defects.
- Received internal IPC-A-610 and J-STD-001 training and applied the standards to PCB assembly and inspection.

TECHNICAL PROJECTS

AutoBOM | C++, Node.js, KiCad

2026

- Extended KiCad's C++ codebase and connected a Node.js backend to build an integrated AI assistant for component sourcing and automated application-circuit generation.
- Turned plain-English requests into in-stock supplier searches, symbol, footprint, and datasheet imports, and missing component-field completion using DigiKey and LCSC data.
- Benchmarked 50 component searches at a 15.8-second median and five application circuits at a 17.4-second median without reusing saved circuit recipes.

UBC Course Scheduler | React, TypeScript, Tailwind CSS

2026

- Built a course scheduling app with React, TypeScript, and Tailwind CSS, using React hooks to manage calendar views and term selection.
- Implemented scheduling logic for recurring and alternate-week classes, with date filtering and duplicate meeting detection.

PickleBot | Autonomous Pickleball-Playing Robot

2026

- Developing an autonomous robot with differential steering and a three-degree-of-freedom swing mechanism.
- Developing real-time computer vision for ball and court tracking on a Raspberry Pi 5 with Hailo-8L and integrating ESP32 control of drive motors and swing actuators.
- Designed a custom KiCad power and control PCB and assembled clamps, shaft couplers, aluminum extrusion, and 3D-printed parts into the drive and swing mechanisms.

TECHNICAL SKILLS

Software: C, C++, Python, Java, Git/GitHub, OpenCV, AI-assisted coding and workflow automation

Embedded Systems: ESP32, ESP-IDF, Arduino IDE, ESP RainMaker, FreeRTOS, Wi-Fi, BLE, UART, I2C, SPI

Hardware & Prototyping: KiCad, Fusion 360, soldering, mechanical assembly, FDM/resin 3D printing